

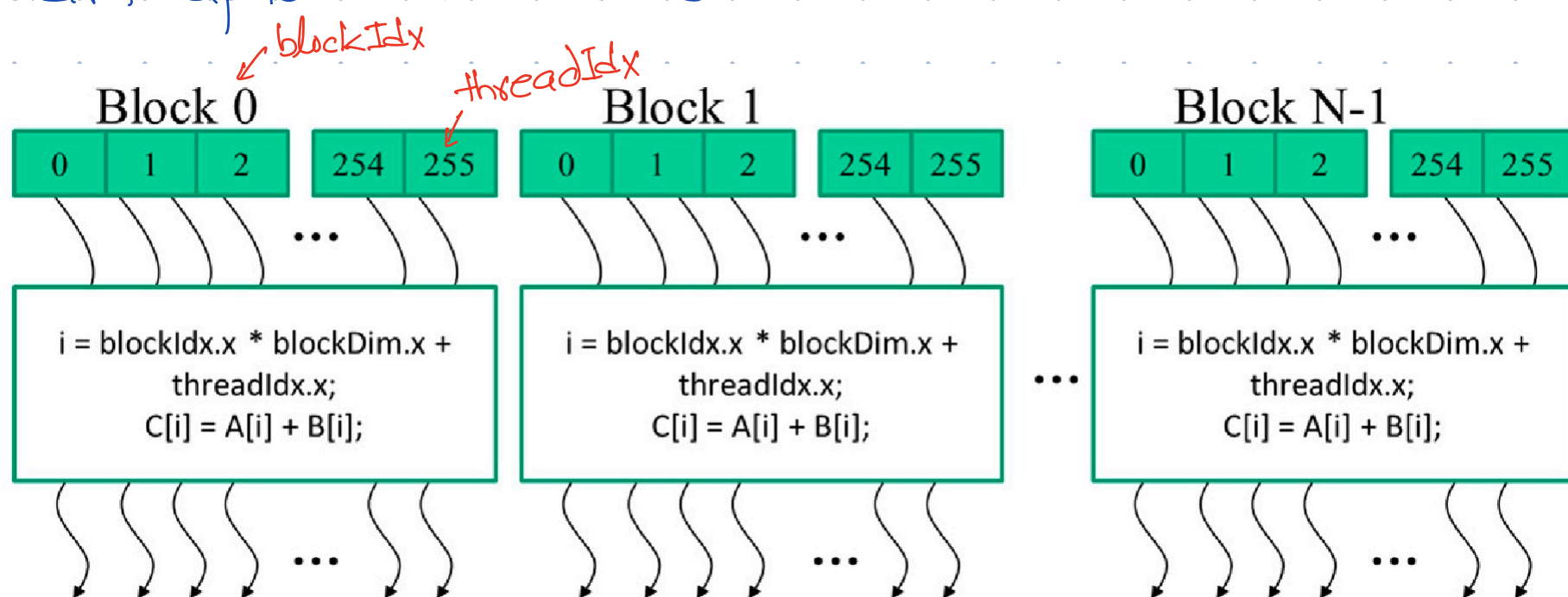
2.5 kernel functions & threading

In Cuda C, a kernel function specifies the code to be executed by all threads during parallel phase.

When a program's host code calls a kernel, the CUDA runtime system launches a grid of threads that are organized into a two-level hierarchy.

Each grid is organized as an array of thread blocks.

All blocks of a grid of same size; each block can contain upto 1024 threads.



2.6 Calling kernel functions

$$\ll \begin{matrix} z, y, x \\ 3, 2, 2 \end{matrix}, \begin{matrix} z, y, x \\ 2, 1, 3 \end{matrix} \gg (\text{args});$$

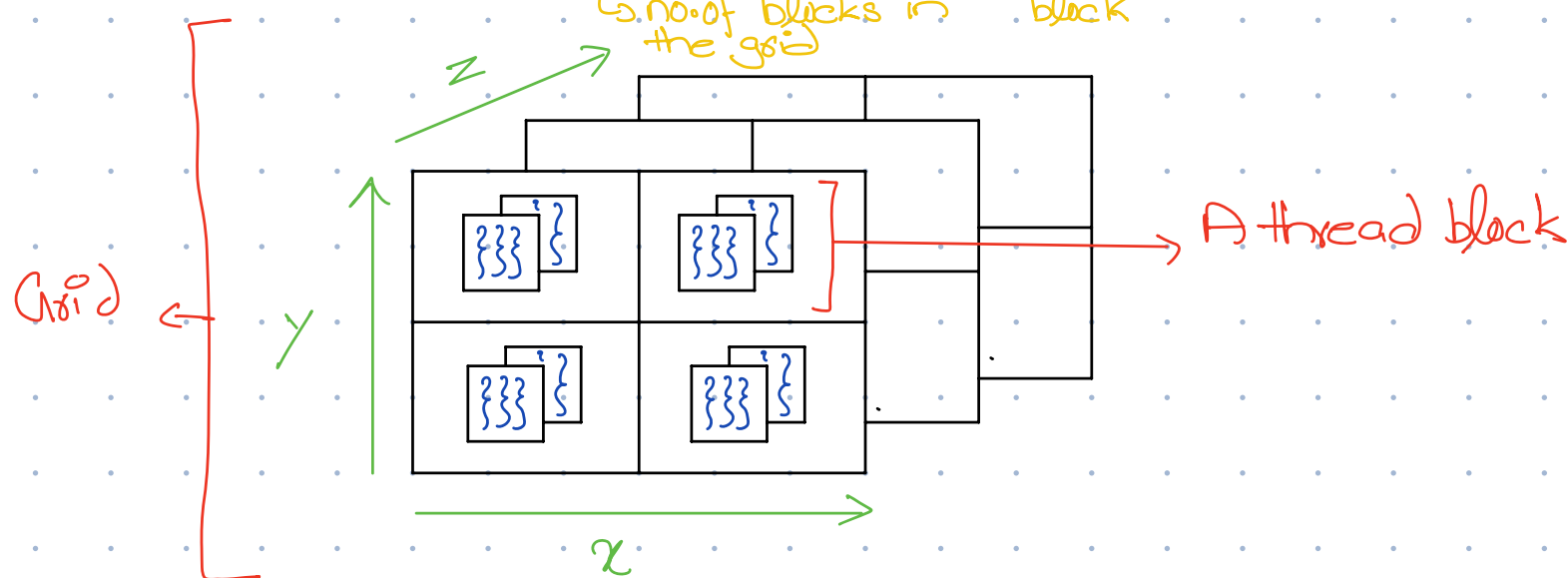
Giới Dìm

black Dinn

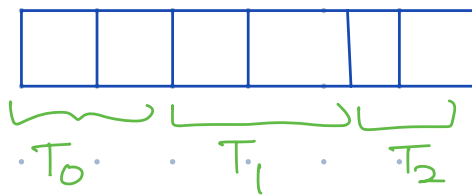
→ no. of threads in each

↳ no. of blue
the grid

• black



block 0



$\ll (1, 1, 3), (1, 1, 6) \gg$

$$i = 0 * 3 + 0$$

$$= 0$$